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Beyond adsorption: Supramolecular bio-gel triggers heavy-metal contaminated soil's self-remediation via electron-cage catalyzed lattice sealing

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Highlights

- Driving soil to spontaneously fix heavy metals is a promising remediation paradigm.
- Electron cage effect drives heavy metal immobilization via isomorphous replacement.
- Spontaneous dealkalization and catalytic nucleation are the key pathways of fixation.

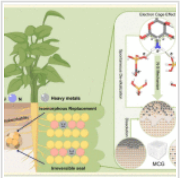
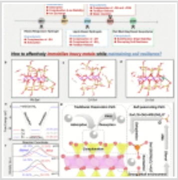
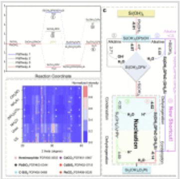
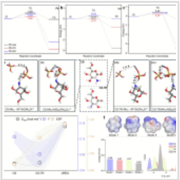
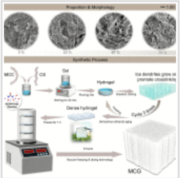
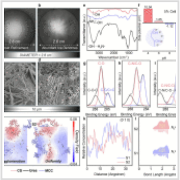
Outline

Abstract

The development of traditional soil remediation agents focuses on the improvement of intrinsic adsorption, leading to the dependence on excessive dosing and environmental disturbance. Herein, we propose a strategy of using alkaline cellulose/chitosan self-assembled gel (MCG) to induce partial dissolution of soil and catalyze heavy metals to participate in soil isomorphous replacement, and seal multiple heavy metals within the soil lattice. The "electron cage" structure of MCG accelerates soil nucleation through dual pathways ("spontaneous de-alkalization" and "catalytic nucleation" of heavy metal silicates), which was verified by a series of characterization techniques. The 90-day soil experiments show that MCG improved soil fertility and reduced the bioavailability of Pb, Zn and Cd by 99.96%, 99.71% and 99.93%, respectively, without harming soil resilience. More importantly, we applied MCG to 50 actual contaminated sites in China, confirming the environmental friendliness and using machine learning for modeling. SHAP analysis revealed that total nitrogen and organic matter content are key evidence for MCG to achieve effective soil remediation through the nucleation-sealing (NS) mechanism. In summary, using degradable hydrogels to activate the intrinsic nucleation mechanism of soil for self-remediation will achieve more environmentally friendly and long-lasting remediation, while overcoming the reliance on material stability.

Graphical abstract

Figures (8)

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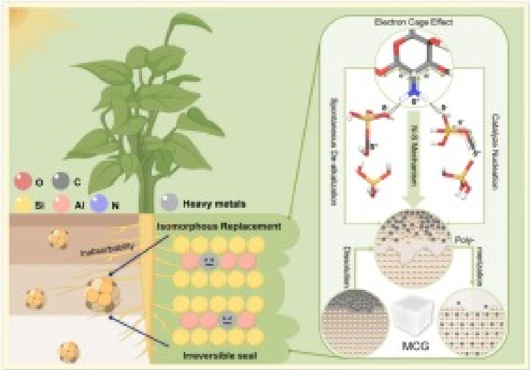

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Extras (1)

Supplementary material

Keywords

Heavy metal; Soil self-remediation; Electron cage effect; Edge nucleation; Chitosan

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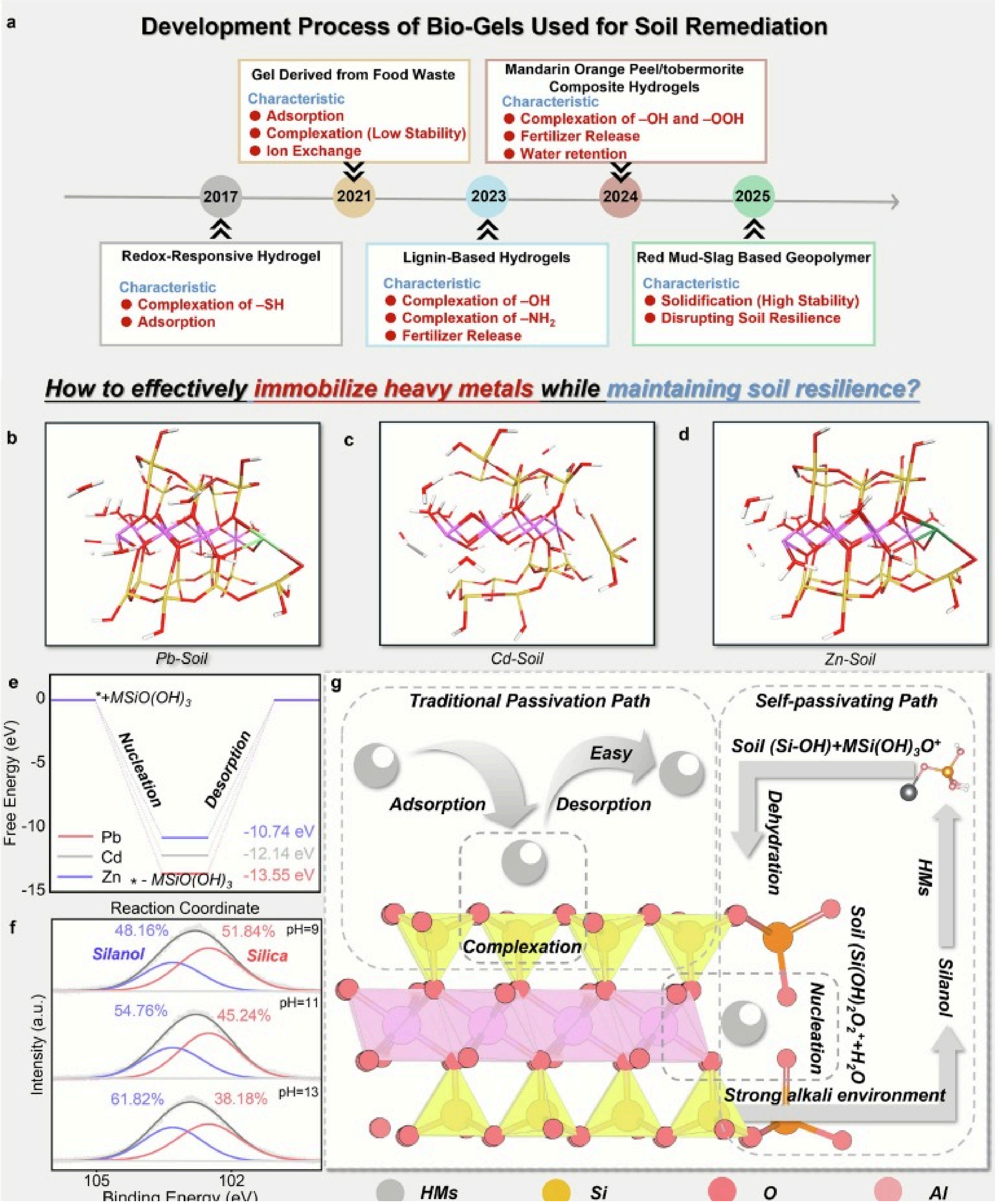
1. Introduction

Soil heavy metal pollution is a worldwide public safety problem [1,2]. In recent years, numerous exogenous remediation materials (biochar, phosphate, metal oxide and metal sulfide, etc.) have been developed to passivate heavy metals in situ contributed by their electron-donating effect [[3], [4], [5]]. In particular, precipitation and adsorption are the most commonly used strategies to control the translocation of heavy metals in soil and groundwater systems [[6], [7], [8], [9]]. Although traditional passivation mechanisms such as complexation and electrostatic attraction can alleviate soil toxicity in the short term, long-term effective passivation depends on the addition of a large amount of remediation materials, which causes serious environmental disturbance and destroys soil resilience

(**Fig. 1**a) [[10], [11], [12], [13], [14], [15], [16], [17]]. Among them, bio-gels have gradually become popular in soil remediation due to their excellent adsorption capacity. However, the mechanism of complexation, adsorption and ion exchange limits the long-term effect of remediation. In recent years, the long-term solidification of heavy metals by geopolymer has been sought after. However, due to the damage to soil resilience, it is difficult to deal with the treatment of the vast majority of heavy metal contaminated farmland. Therefore, it is necessary to develop a material that can effectively immobilize heavy metals while maintaining soil resilience. Based on the regulation mechanism of organic matter on soil mineral evolution, we proposed a strategy of introducing a small amount of functional organic matter into soil to drive soil dissolution and isomorphous replacement to seal heavy metals, which successfully realized a new management paradigm of "material-driven soil self-healing" without destroying soil resilience.



a
Development Process of Bio-Gels Used for Soil Remediation
Gel Derived from Food Waste
Characteristic
Adsorption
Complexation (Low Stability)
Ion Exchange
Mandarin Orange Peel/tobermorite Composite Hydrogels
Characteristic
Complexation of –OH and –OOH
Fertilizer Release
Water retention
2017
2021
2023
2024
2025
Redox-Responsive Hydrogel
Characteristic
Complexation of –SH
Adsorption
Lignin-Based Hydrogels
Characteristic
Complexation of –OH
Complexation of –NH2
Fertilizer Release
Red Mud-Slag Based Geopolymer
Characteristic
Solidification (High Stability)
Disrupting Soil Resilience
How to effectively immobilize heavy metals while maintaining soil resilience?
b
c
d
Pb-Soil
Cd-Soil
Zn-Soil
e
Free Energy (eV)
*+MSiO(OH)3
Nucleation
Desorption
Pb
Cd
Zn
* - MSiO(OH)3
-10.74 eV
-12.14 eV
-13.55 eV
Reaction Coordinate
f
Silanol
Silica
48.16%
51.84%
pH=9
54.76%
45.24%
pH=11
61.82%
38.18%
pH=13
Intensity (a.u.)
105
102
Binding Energy (eV)
g
Traditional Passivation Path
Adsorption
Easy
Desorption
Complexation
Self-passivating Path
Soil (Si-OH)+MSi(OH)3O+
Dehydration
HMs
Soil (Si(OH)2O2++H2O
Silanol
Nucleation
Strong alkali environment
HMs
Si
O
Al

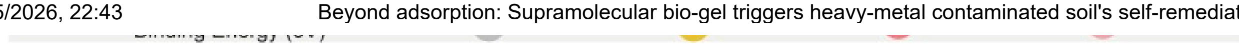


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Fig. 1. **Exploration of the spontaneity of heavy metal nucleation at the soil edge.** **(A)** The development process of the gel used for heavy metal remediation; (**B**—**D**) simulation results of HMS dehydration and condensation at the soil edge; (**E**) Free energy of the nucleation process of HMS at the soil edge (* represents surface); (**F**) XPS spectra of Si *2p* in soil at different pH values; (**G**) Schematic diagram of the NS mechanism for sealing heavy metals.

Soil has intrinsic adsorption ability due to the complexation of endogenous silica-aluminum oxides (such as quartz, plagioclase, etc.) [18,19], but the complexation with a low stability of state results in the easy dissociation of heavy metals [20,21]. However, impurity ions trapped in the soil lattice via the isomorphous replacement (occupying the position of Si^{4+} or Al^{3+} gives the soil permanent negativity) have been shown to be extremely stable. Inspired by this theory, long-term sealing of heavy metals in soil lattice can be achieved using alkaline solutions to dissolve part of soil and induce lattice reconstruction [[22], [23], [24]]. At the soil edge, the dehydration and condensation of heavy metal silicates (HMS) is a key step in isomorphic replacement [25,26]. Although soil can form alkaline heavy metal silicates (AHMS) through the release of silanols and heavy metal hydroxides in alkaline environments, the removal of hydroxyl groups from AHMS (de-alkalization) is necessary step for completing the whole process of soil nucleation [27]. It is known that it is difficult to decrease pH short-timely in soil, which significantly inhibits AHMS to further polymerize [28]. Inspired by the synthesis of ceramics, ammonium-containing substances can significantly increase the ceramic particle size under alkaline conditions, implying that specific types of amino groups may have a yet-to-be-revealed ability to promote spontaneous de-alkalization and further polymerization of AHMS [29]. Through first-principle calculations and characterization techniques, we found that the electron cage structure of chitosan (CS) (“O-N-O" cage structure constructed by the O-hydroxyl group) inhibited the electron transfer of amino group, converting the coordination between amino group and silicate group from covalent interaction to electrostatic force. It drove the spontaneous conversion of AHMS to HMS, enhancing the charge separation of HMS. Finally, it reduced the nuclear barrier and accelerated isomorphous replacement. Therefore, the synthesis of CS-based remediation materials could give the electron cage structures full play to inducing nucleation-sealing and achieve spontaneous de-alkalization and rapid nucleation of AHMS, further driving "soil self-healing" via isomorphous replacement.

However, the weak alkalinity of CS is not sufficient to dissociate the soil, necessitating the introduction of an alkali source such as NaOH. However, the agglomeration of CS due to the deprotonation in a strongly alkaline environment will hinder the exposure of the electron cages [30,31]. The rearrangement effect during freeze-thaw cycles (FTC) can alleviate the aggregation. FTC-based cryogel has high porosity and functional group exposure, which is an ideal form of CS. However, the mechanical strength of CS in NaOH solution is too low to synthesize molding materials. Microcrystalline cellulose (MCC) can enhance the strength of gels by constructing intermolecular hydrogen bond network with CS through supramolecular self-assembly behavior [32]. However, the hydrophobicity caused by intramolecular hydrogen bonds limits the assembly behavior. Interestingly, urea can break MCC intramolecular hydrogen bonds, weaken its hydrophobicity and promote the construction of intermolecular hydrogen bond networks due to the adsorption with the pyran ring [33]. Therefore, we propose to synthesize porous MCC/CS gels (MCG) with surface alkalinity and high electron cage exposure by using supramolecular self-assembly during the process of FTC, without the introduction of any cross-linking agents.

In summary, MCG with alkaline surface, abundant pores and "electron cage" structures was synthesized by FTC with gradient cooling, and it was applied to remediate Pb, Cd and Zn contaminated soils. By monitoring the bioavailability of heavy metals, soil physical and chemical properties, surface phase and bulk phase structure of soil particles within 90 days, it was confirmed that MCG could induce heavy metals and silanol to complete the NS process exponentially through de-alkalizing spontaneously and lowering the nuclear barrier without destroying soil resilience. In the context of the Sustainable Development Goal (SDG 15), the success of MCG in driving the isomorphous replacement effect will provide a breakthrough strategy for developing heavy metal remediation materials since it achieves a long-term sealing of multiple heavy metals within the soil lattice.

2. Methods

2.1. Materials and chemicals

All reagent information is provided in Text S7.

2.2. Synthesis and characterization of MCG

Microcrystalline cellulose/chitosan gel (MCG) with abundant pores was synthesized by freeze-thaw cycling process (FTC) in deep eutectic solvent (7% NaOH/12% urea/81% H_2O) containing microcrystalline cellulose and chitosan. Solubility is a function of the degree of cross-linking. The effect of the pre-freezing time of urea- NaOH solution (20, 30, and 40 min, respectively) on the dissolution of cellulose and chitosan was explored at −50 °C (**Fig. S1**), and 30 min was used as the pre-freezing time of the deep eutectic solvent (DES). In addition, the cooling rate will significantly affect the subcooling degree of the solution and then affect the nucleation process of ice crystals. Therefore, under the optimal solubility, the freezing procedure was performed using two schemes: flash freezing (−50 °C) and gradient cooling (cooling rate: $-10\,°C\cdot h^{-1}$), and the pores were compared.

5 g microcrystalline cellulose (MCC) and 2 g chitosan (CS) were dissolved in precooled 7% NaOH/12% urea/81% H_2O solution for dispersion and activation, respectively, to obtain MCC and CS sols. The two sols were mixed at the ratios of 0:1 (0%), 1:1 (50%), 2:1 (67%) and 3:1 (75%), and two groups of samples were prepared in the same way. After stirring (150 rpm) for 30 min, one group was placed in a flash freeze at −50 °C for 12 h (the other group was subjected to gradient cooling at a rate of $-10\,°C\cdot h^{-1}$), thawed at room

temperature for 20 min, and such freeze-thaw cycles were performed three times. After defoaming treatment, deionized water was added to the wash, and the cured gel was transferred to absolute ethanol for solvent replacement for 48 h. After freezing at −50 °C for 6 h, the gel was freeze-dried at −60 °C±5 °C for 24 h to obtain MCG (**Fig. S2**).

Scanning electron microscopy (SEM) was used to compare the pore structure of gels prepared by different processes. To further reveal the supramolecular self-assembly mechanism, an energy spectrometer (EDS), X-ray photoelectron spectroscopy (XPS) and Fourier transform infrared spectroscopy (FT-IR) were used to characterize the surface element distribution and chemical properties of MCG and further analyze the zeta-potential of MCC. The specific instrument information is provided in **Text S1**.

2.3. Molecular dynamics simulation

To further explore the role of each component (microcrystalline cellulose, chitosan, sodium hydroxide, urea) in the self-assembly process, an explicit solvation model of cellulose/chitosan sol was constructed using the Amorphous Cell module in Materials Studio. The Forcite tool was used to perform geometric structure optimization and molecular dynamics (MD) simulation, record mean square displacement (MSD), relative concentration of CS, field density of N of CS, and diffusion coefficient of different equilibrium states of the system and use the script (**Text S2**) to calculate the length and number of hydrogen bonds. The specific calculation process is shown in **Text S3**.

2.4. Density functional theory calculations

All the calculation schemes are based on density functional theory in DMol^3 in Materials Studio 2017. Free energy ($\Delta G_{298.15\ K}$) calculations (**Text S4**) of the nucleation process were performed using a soil model based on a representative montmorillonite lamellae structure (**Fig. S3**). A combination of generalized gradient approximation (GGA), Perdew-Burke-Ernzerhof (PBE), and TS method was used to accurately describe the weak interactions in the density functional theory (DFT) calculations in the implicit solvation model (dielectric constant=78.54). Further calculation details are provided in **Table S2**.

2.5. Preparation, characterization, and detection of soil samples

Based on the plum blossom sampling method, all soil samples were collected from the soil layer A of the Popular Science Botanical Garden (North latitude: 41°5′15″, east longitude: 121°6′59″), which were air-dried for 24h and sieved to ensure that all soil particle sizes were less than 2mm. The original soil properties are shown in **Table S3**, and the detection methods for each item are shown in **Text S5**. To reflect the application value of MCG, according to the *Soil environmental quality Risk control standard for soil contamination of agricultural land* (GB 15618–2018), 0.1M lead nitrate, 0.1M zinc nitrate, and 0.1M cadmium nitrate were prepared and added to 1 Kg pretreated clean soil. The initial concentrations (C_0) of Pb, Zn, and Cd in the soil were adjusted to 240ppm, 500ppm, and 200ppm, respectively.

Effect of pH on the mineral composition of soil: 1M NaOH and 0.1M HCl were prepared, and the pH of three parts of the soil was adjusted to 9,11,13, respectively. After stirring, the soil was placed in a cool place to air dry for 2days, and the chemical morphology of silicon was characterized by XPS after grinding.

Effects of different amines on the nucleation of alkaline soil: According to HJ 962–2018, 5 groups of 10g soil samples were dissolved in 25mL NaOH solution (pH=13). After full stirring, 1.00g of hydroxylamine hydrochloride (ClH_4NO), ammonium acetate (NH_4Ac), ammonium carbonate ($(NH_4)_2CO_3$), urea ($CO(NH_2)_2$) and CS were added, respectively. After stirring, 5mL of 1M acetic acid was dropped to block the polymerization process of silicate and then filtered and dried at 60°C. After grinding, XRD was used to characterize the crystalline phase, and MDI Jade 6 was used to resolve the crystalline phase composition.

Monitoring of crystal characteristics in soil culture experiment: While monitoring soil pH, soil samples with pH=13, 12, 11, 10, and 9 were taken and thoroughly ground. XRD and FT-IR were used to detect the heavy metal morphology of the bulk phase and surface phase. To verify the distribution of heavy metals in soil lattice, the presence and content of different heavy metals in soil samples 90 days after restoration were characterized by etch-XPS.

In addition, the passivation efficiency of soil heavy metals obtained by the calcium chloride extraction method (**Text S6**), which can represent the remediation effect, can be obtained from **Eq. 1**:

$$\text{Passivation efficiency}(\%) = \frac{C_0 - C}{C_0} \times 100\% \quad (1)$$

Where C_0 is the initial concentration and C is the current concentration.

2.6. Statistical analysis

One-way ANOVA analysis and Tukey's test ($p<0.05$) were employed to compare significance between treatments. All experiments were performed in triplicate, and the results were presented as mean values ± SDs ($n=3$). To make the FT-IR spectra of different samples comparable, we performed normalization of the test results using **Eq. 2**. The statistical analysis was performed using IBM SPSS statistics 26. Subsequently, each was averaged and plotted using Origin 2022.

$$X' = \frac{x - \min(x)}{\max(x) - \min(x)} \quad (2)$$

where X' is the normalized result, x is the data, $\max(x)$ is the maximum value, and $\min(x)$ is the minimum value.

3. Results

Traditional soil heavy metal remediation materials rely on the kinetics and stability of adsorption process, and their remediation effect is seriously limited by the lifetime of materials. How to decouple the correlation between remediation effect and material stability is the primary problem in this field. This study found that a chitosan-based gel, via electrostatic effects, catalytically alters silicate crystallization direction. Accordingly, a degradable bio-gel was designed for driving soil to "remediate itself" by inducing heavy metals to participate in edge nucleation, thereby sealing heavy metals within the soil lattice irreversibly and overcoming the dependence of the remediation effect on material stability.

3.1. Feasibility of edge nucleation of heavy metals

Based on the isomorphous replacement effect, driving heavy metals to participate in soil nucleation can stably seal them in the soil lattice [34]. However, the nucleation process under natural conditions requires thousands to tens of thousands of years [35,36]. Understanding the thermodynamics and spontaneity of heterogeneous heavy metal nucleation at soil edges can provide the theoretical basis for utilizing isomorphous replacement [25,26]. The edge nucleation processes of three heavy metal (Pb, Zn, Cd) silicates under neutral conditions were simulated using first-principles calculations using a representative co-edge octahedral lamellae connecting two co-angular tetrahedral lamellae and forming a 2:1 layered structure (Fig. 1B-D) [25,26]. After zero-point vibration energy correction, the free energies ($\Delta G_{298.15\ K}$) of the three HMS nucleation processes are −10.74, −13.55, and−12.14eV respectively (Fig. 1E), indicating that HMS is the raw material required by heavy metals to participate in the edge nucleation to complete the NS mechanism, and the sealing process shows thermodynamic irreversibility [37]. However, silicon in soil often exists as a stable crystalline silica (SiO_2) structure, which makes it difficult to form sufficient HMS under natural conditions. Inspired by the glass synthesis process, we speculate that the introduction of alkaline substances into soil dissolves the soil surface lattice and upon contact with heavy metals will significantly increase the content of HMS [[38], [39], [40]]. To test this hypothesis, soil samples with different pH (9–13) were prepared and XPS was used to detect the transformation of silicon oxide forms at different pHs (Fig. 1F). The results showed that with the increase of pH, SiO_2 gradually dissociates and converts to silanol (48.16% to 61.82%), which will provide sufficient substrate for the formation of HMS [31,41,42]. Therefore, we proposed the strategy of using alkali to dissolve soil to form silanol, which generated HMS. Subsequently, it further dehydrates and nucleates at the soil edge, changing the passivation mechanism from complexation into NS (Fig. 1G). However, the detailed reaction mechanism has not been elucidated, that is heavy metals will be hydroxylated under alkaline conditions and then whether it can react with silanol to form HMS, further completing the NS procedure.

3.2. Potential nucleation pathways at the soil edge

Clarifying the reaction pathways of the nucleation-sealing (NS) mechanism at different pH levels is conducive to understanding and regulating soil evolution and nucleation at the atomic level. Based on the similarities shown in Fig. 1D, taking Pb(II) as an example, we propose four potential reaction paths of the NS mechanism between silanol and heavy metals using first-principles calculations (Fig. 2A) (The detailed reaction process is shown in **Eq. S2-S17**) [37,43]. Although carbonate is a common form in neutral/weak alkaline soils, heavy metals will be converted into hydroxide precipitates with lower solubility products in an environment where alkalinity continues to increase [44,45]. According to **Eq. S2**, the Gibbs free energy ($\Delta G_{298.15\ K}$) of dehydration and condensation of silanol and heavy metal hydroxide (MOH) to form AHMS ($Si(OH)_3OMOH$) is −0.01 eV at 298.15 K, confirming the spontaneous dehydration and polymerization of silanol and MOH. However, previous studies have shown that the formation of HMS ($Si(OH)_3OMO(OH)_3Si$) is thermally hindered in alkaline conditions, indicating that there is a yet-to-be-revealed thermodynamic inhibition step during HMS formation [28]. Accordingly, we calculated $\Delta G_{298.15\ K}$ for two possible pathways (**Pathway 1 and 2 in** Fig. 2A): 1. direct condensation of AHMS-silanol to HMS (**Eq. S4**); 2. the further formation of HMS by AHMS with silanol after OH^- removal (**Eq. S5**) in an alkaline environment. The calculated results show that neither of the two pathways can be carried out spontaneously ($\Delta G_{298.15\ K}$ are 6.69 eV and 2.71 eV, respectively). It can be concluded that the spontaneous de-alkalization of HMS (the conversion of $Si(OH)_3OMOH$ to $Si(OH)_3OMO(OH)_3Si$) is one of the main bottlenecks in generating HMS in alkaline environments. However, when the pH is reduced, the participation of H^+ causes the spontaneous removal of OH^- from $Si(OH)_3OMOH$ ($\Delta G_{298.15\ K}$=-4.97 eV) to form $Si(OH)_3OM^+$ (**Eq. S11**), which facilitates further reaction with silanol to form HMS (**Eq. S12**, $\Delta G_{298.15\ K}$=−0.35 eV). The above calculation results imply that, although HMS can be formed by alkali dissolution at a wide time scale, it depends on the slow decline of pH or the assistance of exogenous buffer to complete the secondary crystallization. However, this slow process is unsuitable for the emergency remediation of heavy metals in the field, such as in contaminated farmland near mining areas, urban brownfields, and abandoned industrial sites [46,47]. Therefore, how to promote the nucleation of $Si(OH)_3OPbOH$ under alkaline conditions is an urgent problem to be solved to achieve rapid nucleation and seal in soil.

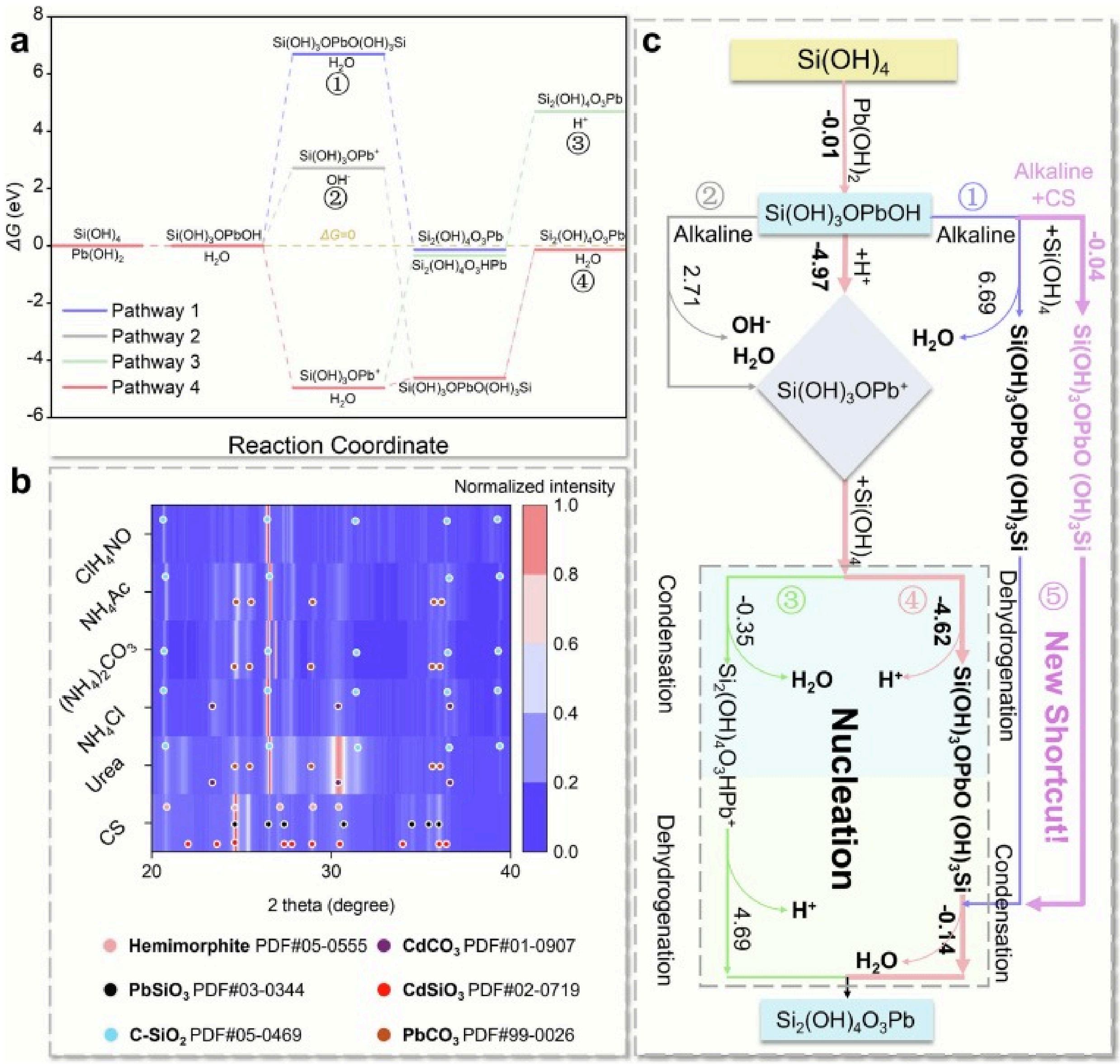

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Fig. 2. **Catalytic effect of chitosan on the de-alkalization of heavy metal silicates.** (**A**) Free energy curves of the reaction between heavy metals and silanols at different pH

values; (**B**) XRD spectra of reaction products of different ammonia-containing substances with soil alkali solution; (**C**) Detailed reaction paths of heavy metals and silanols at different pH and new shortcut opened by chitosan.

3.3. Pathway of spontaneous de-alkalization

Inspired by ceramic synthesis, amino groups can significantly improve the particle size of ceramics under weak alkaline conditions [29]. We hypothesize that the introduction of ammonia-containing substances into alkali-soluble soils may facilitate the nucleation process of HMS. Therefore, we tried to introduce a variety of ammonia-containing substances (hydroxylamine hydrochloride, ammonium acetate, ammonium carbonate, ammonium chloride, urea and CS) into the alkali-dissolved heavy metal contaminated soil (silanol). And after full mixing, acetic acid was used to block the dehydration condensation of HMS (to avoid the secondary generation of SiO_2), and the obtained solid products were collected [48]. Analysis on the products of the above five treatments using XRD showed that the products of the first four treatments were mainly composed of crystalline silica and heavy metal carbonate. Encouragingly, the CS treatment group formed a large amount of HMS in a short period of time (Fig. 2B), suggesting that the catalytic activity of the amino group was due to the potential structure-activity relationship. Based on this, we added the amino group of CS to the first principles calculation and calculated the $\Delta G_{298.15\ K}$ of AHMS converted to HMS. The results show that under the catalytic action of CS, $Si(OH)_3OPbOH$ can spontaneously condense with silanol to HMS (**Eq. S18-S20**) under alkaline conditions, successfully constructing a new pathway of spontaneous de-alkalization of HMS with fewer steps in alkaline environment (Fig. 2C). It breaks through the main barrier that AHMS is not able to spontaneously de-alkalization in an alkaline environment.

3.4. Accelerated soil nucleation dynamics

According to the soil isomorphous replacement effect, the formation of new lattice in soil still requires several years of evolution even under neutral conditions, indicating that in addition to the spontaneous transformation of AHMS to HMS, the further nucleation process of HMS is still hindered by high energy barriers [49]. According to Fig. 2C, nucleation is mainly carried out through **path 4**, because **path 3** is limited by the non-spontaneous dehydrogenation step. Therefore, we speculate that the slow nucleation dynamics of soil come from the binding step of $Si(OH)_3OM^+$ to silanol. To explore the effect of CS on this process, we further calculated the effect of the amino group and hydroxyl group in CS participating in the HMS formation process of $Si(OH)_3OM^+$ according to the transition state theory (Fig. 3A-C) [50]. In the system without CS, the silicate formation of the three heavy metals had to overcome a huge energy barrier (Pb: 0.29 eV, Cd: 0.27 eV, Zn: 0.42 eV). At the same time, the hydroxyl group in CS had no obvious catalytic effect (Pb: 0.24 eV, Cd: 0.34 eV, Zn: 0.36 eV), while the amino group on the β-C of CS significantly reduced the formed barrier of HMS (Pb: 0.08 eV, Cd: −0.01 eV, Zn: 0.24 eV), indicating that amino group was the active center for HMS formation catalyzed by CS. According to the Arrhenius equation (**Eq. S20**), the reaction rate constants (k) of the nucleation process before and after CS participation were calculated. The results showed that the rate constant (k) for Pb, Cd, and Zn nucleation increased by approximately 3.4×10^3, 5.1×10^4, and 1.1×10^3 times, respectively, upon catalysis by the amino group in CS [51]. It indicated that CS not only constructed a new path for spontaneous de-alkalization of HMS but also reduced the energy barrier of the binding between $Si(OH)_3OM^+$ and silanol, further accelerating NS mechanism. It can be concluded that it breaks through the slow nucleation dynamics of the isomorphous replacement effect. However, the reason for the special catalytic activity of CS has not been revealed, compared with other ammonia-containing substances.

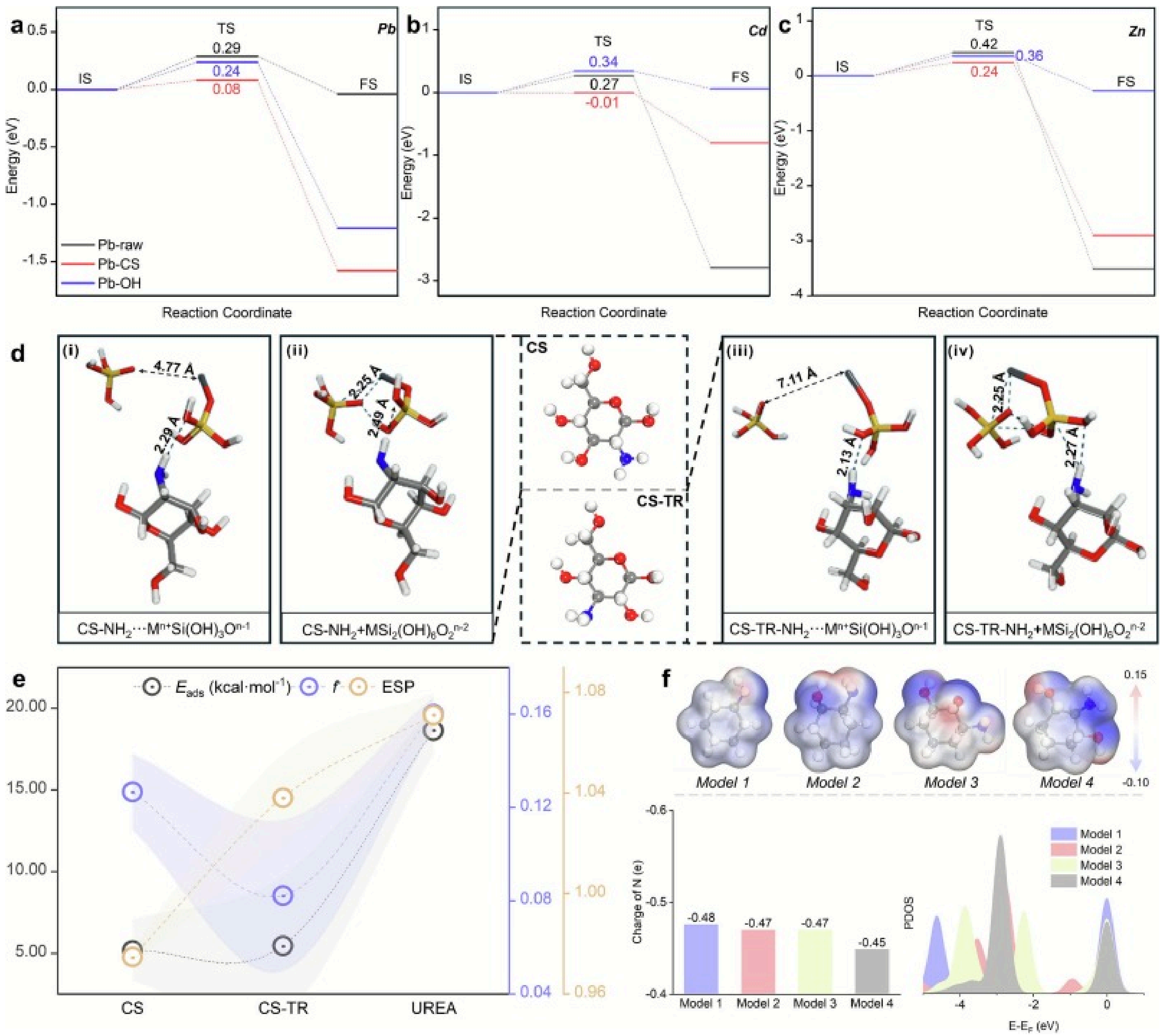

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Fig. 3. **Special structure-activity relationship between the hydroxyl group and amino group.** (**A-C**) Form-nuclear barrier of heavy metal silicate under the action of different functional groups of CS; (**D**) Adsorption configuration of $Si(OH)_3OM^+$ by CS and its isomers. (i, iii) configurations before adsorption, (ii, iv) configurations after adsorption;

(**E**) Adsorption energies, Fukui functions, and electrostatic potential charges of amino groups in urea, CS and its isomers; (**F**) Structure-function relationship of amino electrostatic potential, charge amount, and projected density of states with different oxygen atom distributions.

3.5. Revelation of the electron cage effect

To further reveal the structure-activity relationship between the amino microstructure (O-N-O) in CS and its catalytic activity for silicate ($Si(OH)_3OM^+$) nucleation, the properties of CS and its isomer (CS-TR), the electrostatic potential charge (ESP), adsorption energy (E_{ads}), electrophilic index (f^-) and the transition state energy barrier (E_a) of HMS nucleation in urea were calculated precisely [37,52,53]. The results showed (**Fig. S4**) that the E_a of CS, CS-TR, and urea treatment groups were 0.08, 0.21, and 0.48 eV respectively, showing significant differences in catalytic performance. Among them, the coordination bond length of HMS and CS-TR (2.27 Å) is significantly smaller than that of CS (2.49 Å) (Fig. 3D). The strong Lewis alkalinity of CS-TR leads to excessive coordination of silicates, forming an overly stable intermediate state, which hinders the lattice reconstruction of HMS. However, CS has moderate adsorption energy for $Si(OH)_3OM^+$, which can avoid the excessive allocation of positive charge by silicate and enhance the nucleation ability of $Si(OH)_3OM^+$. Further analysis on the charges, E_{ads}, and f^- of amino groups on CS, CS-TR, and urea (Fig. 3E) shows that the amino group on β-C of CS has a strong electron donor tendency, a low negative charge and E_{ads}, which avoids the inhibition of silicate nucleation by excessive coordination. According to the soft and hard acid and base theory, the above phenomenon can be attributed to the low electronegativity of the N atom (3.04) relative to the O atom (3.44), which leads to the induction of the amino group by the ortho-hydroxyl groups, which limits the transfer of electrons and strengthens the hard base properties of the amino group on the β-C of CS [54].

Similarly, in **Fig. S5**, CS causes metal ions in $Si(OH)_3OM^+$ to collect more positive charges, which means that CS leads to stronger polarization and enhanced charge separation at both ends near and away from silanol, resulting in easier nucleation of $Si(OH)_3OM^+$ with silanol. We further calculate the PDOS of N *2p* and its charge population in different structures by using four models shown in Fig. 3F and discuss the effect of O atom's distributions on the electronic structure of the amino group in detail. DOS at the edge of the Fermi level determines the concentration of electrons that can participate in chemical reactions [55]. Among them, the first local peak of N *2p* electrons has the strongest de-localize phenomenon, which promotes electron migration to lower energy levels and increases the difficulty of electron transfer, reflecting the “O-N-O" cage structure constructed by amino group and 2 o-hydroxyl groups in CS, which imparts a strong negative electric atmosphere to the amino group and induces the enhancement of local electrophilicity of heavy metal ions, while restraining electron transfer. The coordination between CS and silicate (CS-$Si(OH)_3OM^+$) is transformed from covalent to electrostatic force, which is referred to as the “ONO electron cage” effect. A more detailed explanation of the mechanism of the electronic cage effect can be found in Text S8.

In summary, different from the traditional soil remediation mechanisms (complexation, ion exchange, electrostatic attraction, and precipitation) [5], based on the “ONO” electron cage effect, the development of alkaline materials containing CS and their application in heavy metal-contaminated soil can be achieved through two pathways (1. A new pathway for HMS spontaneous alkalization was constructed; 2. Reducing the nuclear barrier of $Si(OH)_3OM^+$ and silanol) to initiate and catalyze the soil NS mechanism and induce the irreversible fixation of heavy metal ions by the soil itself through the isomorphous replacement.

3.6. Synthesis of basic chitosan-based gels for inducing NS mechanism

To fully realize the induction of intrinsic soil NS mechanism by the ONO electron cage structure and minimize environmental disturbance, it is urgent to develop a carrier with high surface alkalinity, aldehyde crosslinker free, rapid degradation, and high porosity. Interestingly, we have proposed a novel scheme to produce MCC/CS gel (MCG) using microcrystalline cellulose as a carrier, which meets all the above requirements by inducing supramolecular self-assembly behavior in a deep eutectic solvent (urea-NaOH system) by freeze-thaw cycling (FTC) with gradient cooling. The molecular composition will significantly affect the degree of self-assembly of the gel [56,57]. When MCC sol volume/CS sol volume=67%, the resulting MCG exhibited the most abundant pore structure (Fig. 4A). Accordingly, the synthesis of MCG was performed using the protocol shown in Fig. 4B.

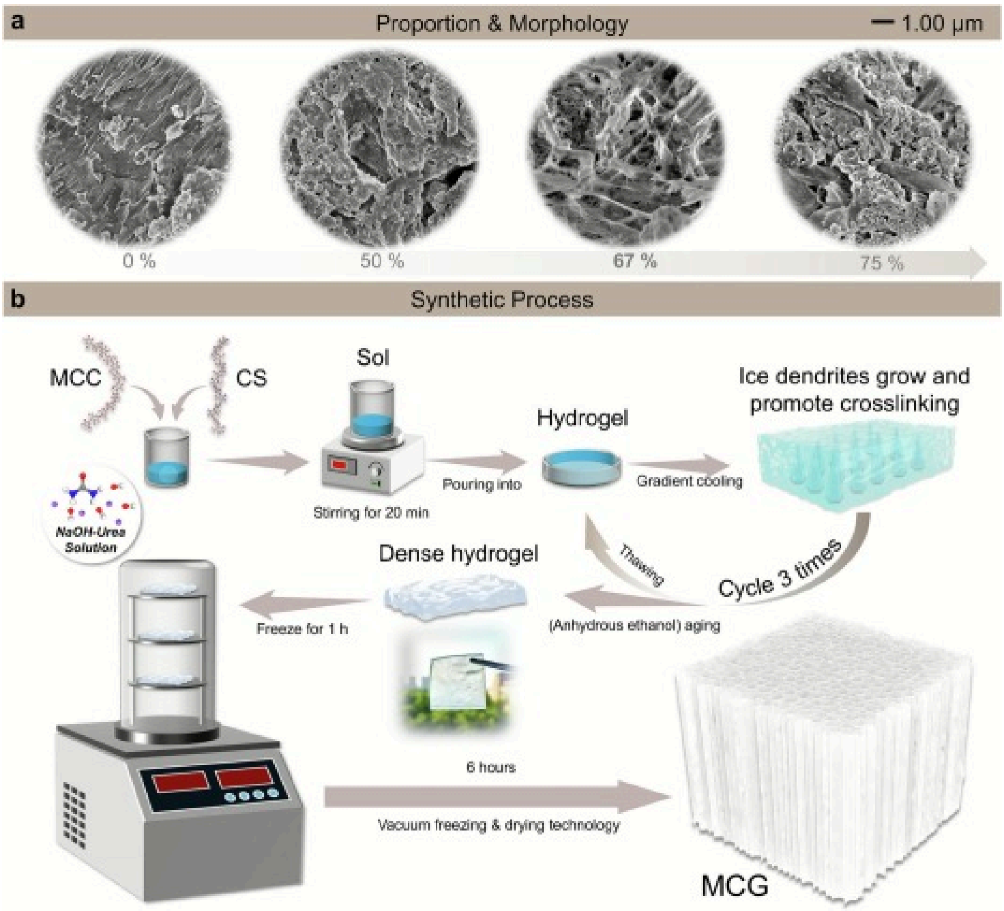

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Fig. 4. **Synthesis of supramolecular self-assembled biogels.** (**A-C**) Morphologies of gels obtained by different volume fractions of CS sol. (**D**) Preparation of self-assembled supramolecular bio-gels from MCC and CS via gradient cooling freeze-thaw cycling.

3.7. MCG performance characterization and self-assembly behavior analysis

To evaluate the potential of MCG to induce the NS mechanism, the surface morphology and properties of MCG were observed and characterized. As shown in Fig. 5A-D, compared with the sharp freezing treatment, the gradient cooling process can inhibit the occurrence of the undercooling effect, which is conducive to the formation of coarser ice crystals. After freeze-drying, a more abundant pore structure is obtained, and the active amino group of CS is fully exposed [58]. FT-IR spectra (Fig. 5E) and XPS spectra (Fig. 5G-I) of MCC, CS, and MCG indicate that the deep eutectic solvent activates -OH, $-NH_2$, and other functional groups by disrupting the intramolecular association of MCC with CS [[59], [60], [61], [62], [63]]. After FTC, the active groups undergo sufficient supramolecular self-assembly through intermolecular hydrogen bonding, which is conducive to the formation of abundant pore structure and exposure of abundant active amino groups, promoting the enrichment of heavy metal ions, and realizing the “spontaneous de-alkalization” and “catalytic nucleation” of HMS. Among them, MCC and CS are entangled with each other to induce active groups to construct an intermolecular hydrogen bond network, and the self-assembled supramolecular structure can be constructed without chemical cross-linking [56]. MCG is given excellent degradability and has lower environmental disturbance while fully exposing the “ONO electron cage”. Notably, the zeta-potential measurements (Fig. 5F) showed that MCC was significantly surface-negative under synthetic conditions. Meanwhile, according to previous research [64], the CS similarly carried a negative charge above pH = 10 due to the deprotonation of the hydroxyl group. This would cause CS and MCC to agglomerate separately due to electrostatic repulsion, and the active amino group would not be fully exposed (**Fig. S6**). However, according to the EDS mappings of MCG before and after FTC (**Fig. S7**), the distribution of O and N elements on the sample surface after FTC is more uniform, and the proportion of N elements increases from 21.32% to 27.95%, indicating that FTC promotes the dispersion of CS and is more conducive to the exposure of the “ONO electron cage” structures.

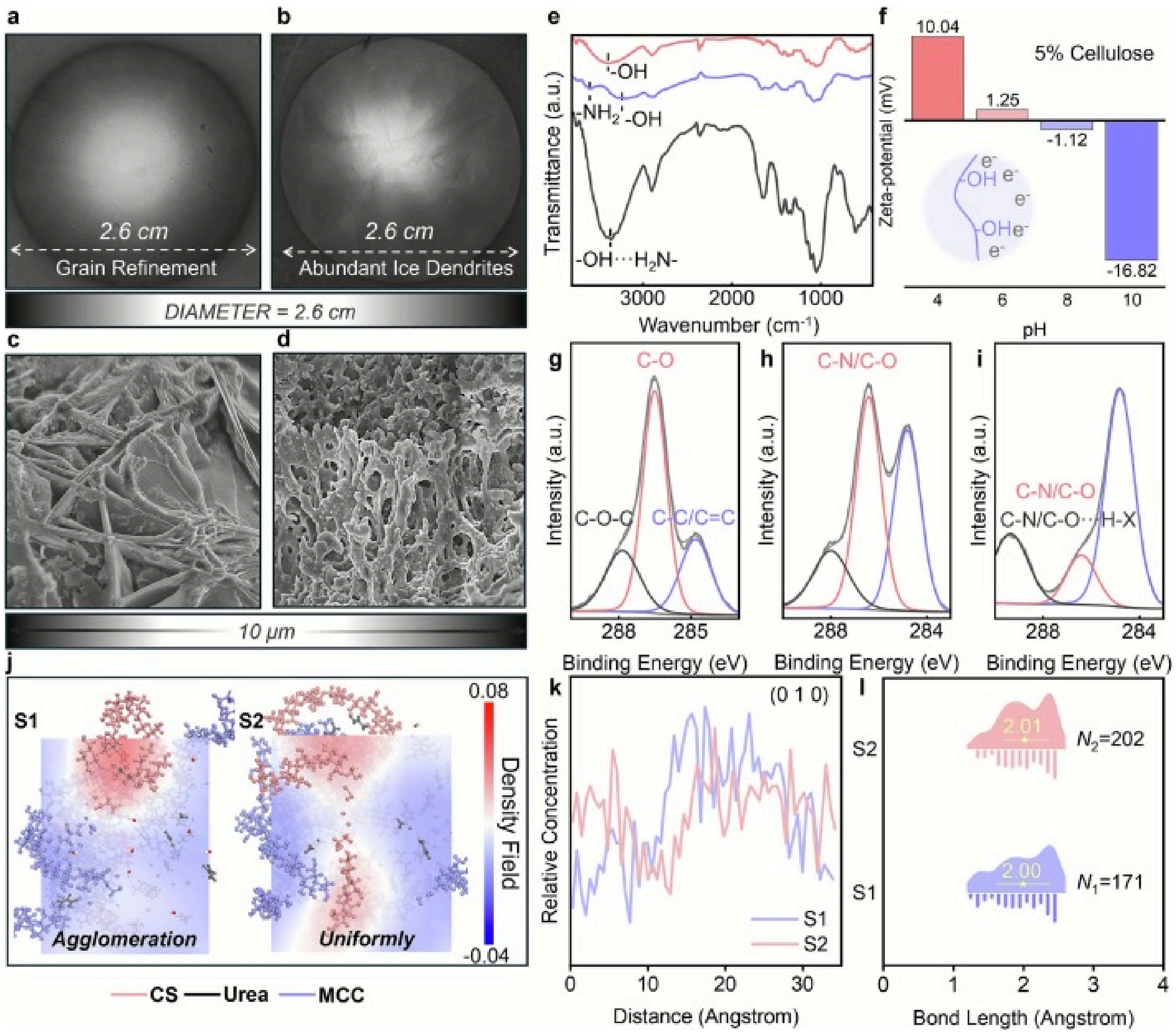

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Fig. 5. **Hierarchical structure and assembly behavior of supramolecular biogels.** The macroscopic morphology of gels under (**A**) freezing treatment and (**B**) gradient cooling. Microstructure of gels obtained by (**C**) freezing treatment and (**D**) gradient cooling; (**E**) FT-IR spectra of MCC, CS and MCG; (**F**) Zeta-potential of MCC; C 1s XPS spectra of (**G**) MCC,

(**H**) CS and (**I**) MCG; (**J**) Color maps of density field profile of CS, (**K**) relative concentration distribution of CS and (**L**) number of hydrogen bonds of gel models obtained by direct freezing and freeze-thaw cycles.

To further observe the internal driving force and assembly mechanism of CS in FTC, MD was used to simulate the kinetic behavior of CS and MCC molecules before and after FTC in water (The model is shown in **Fig. S8**) [37]. The density field distribution of N in CS, the relative concentration of CS, and the number of hydrogen bonds in different systems were counted. The color filling plot of the density field profile shown in Fig. 5J shows that CS is uniformly distributed during freeze-thaw, which is consistent with the relative concentration distribution of CS along the direction of (010) (Fig. 5K), which benefits from the rearrangement effect during the melting process. At the same time, the increase in the average length of hydrogen bonds reflects the conversion of intramolecular to intermolecular hydrogen bonds, and the increase in the number of hydrogen bonds verifies the facilitation of FTC procedure in constructing self-assembled supramolecular networks and exposing the electron cage structures(Fig. 5L) [65]. In summary, MCG is an ideal inducer of soil NS mechanism due to its ability to dissolve soil to form HMS and catalyze the occurrence of isomorphous replacement effect due to its cleanliness, degradability, surface alkalinity, and high “ONO” structure exposure.

3.8. Soil remediation experiments and speciation monitoring of heavy metals

To evaluate the actual sealing effect of MCG on soil heavy metals through the “electron cage” effect induced-isomorphous replacement, the application experiments were carried out in the soil contaminated with Pb (240 ppm), Zn (500 ppm), and Cd (200 ppm) according to the *Soil environmental quality Risk control standard for soil contamination of agricultural land* (GB 15618–2018). And the bioavailable heavy metal content was monitored by the $CaCl_2$-extraction method combined with FAAS, and the passivation efficiency was obtained by **Eq. 1**. Considering the complexity of the soil environment, we systematically investigated the remediation effects at different dosages (0 to 0.20 g·g soil^{-1}), moisture (0 to 100%), and temperature (−10 to 30 °C). After 5 days of culture, the passivation efficiencies of different groups were significantly different [37]. The results showed that the optimal dosage of MCG was 0.1 g·g soil^{-1} (Fig. 6A). The optimal moisture was 50% (Fig. 6B), and the increase in moisture promoted the occurrence of the soil decomposition process causing by expanding the alkali solubility range. Meanwhile, the formation of HMS was promoted by accelerating pH neutralization (**Path 4 in** Fig. 2C). In addition, although the optimal temperature was 10 °C (Fig. 6C), the passivation efficiency at other temperatures did not show a significant reduction, reflecting the strong temperature adaptability of MCG to achieve irreversible fixation of heavy metals through the NS mechanism.

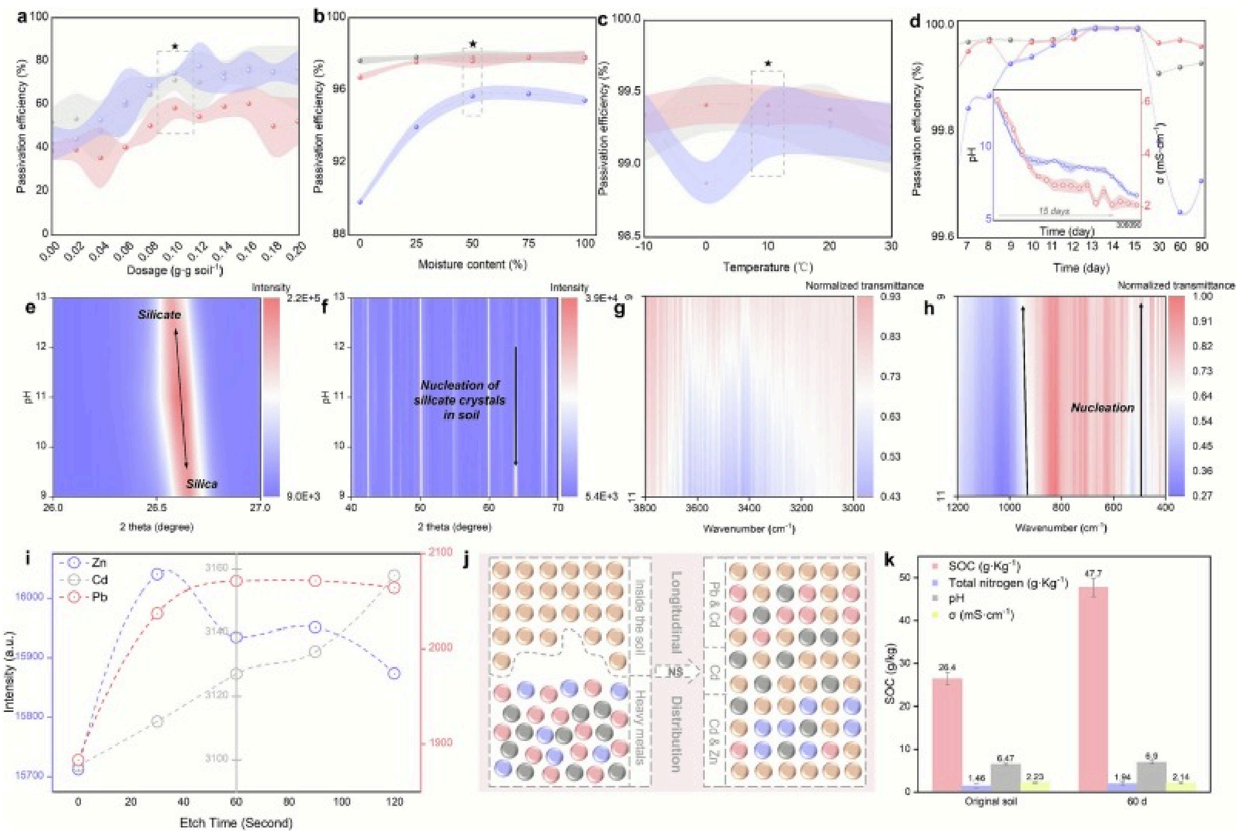

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Fig. 6. **Soil remediation capacity and environmental impact of supramolecular biogels.** Passivation efficiency of heavy metal at different (**A**) dosage, (**B**) moisture content, and (**C**) temperature (Blue: Zn; Red: Pb; Gray: Cd); (**D**) Relationship between passivation efficiency, pH, conductivity, and time; After the remediation experiment, (**E, F**) XRD spectra and (**G, H**) normalized FT-IR spectra of soil samples collected at different pH values. (**I**) The relationship between different etching times and heavy metal content and (**J**) its distribution diagram of soil samples collected on the 90th day after restoration; (**K**) SOC, TN, pH, and electrical conductivity of the original soil and 60days after restoration.

More importantly, to verify the long-term stability of the NS mechanism, a 90-day incubation experiment was performed under the optimal application conditions, with simultaneous monitoring of bioavailable heavy metal content, pH, electrical conductivity (σ), crystalline phase composition, normalized FT-IR spectrum and particle size distribution. The passivation efficiency of the three heavy metals gradually increased and reached its peak (99.99%) on the 13th day. At the same time, σ gradually decreased and returned to the original soil level on day 13 ($2.15 \pm 0.20\ mS \cdot cm^{-1}$, $\sigma_{original\ soil}$: $2.23 \pm 0.14\ mS \cdot cm^{-1}$), indicating that MCG effectively reduced the migration ability of heavy metals in the contaminated soil (Fig. 6D) [66]. In addition, we sampled the soil at different pH levels after testing the soil pH and analyzed the speciation of heavy metals in the bulk phase of the soil samples using XRD. When pH=13, HMS formed rapidly. With a decreasing pH and the catalytic effect of MCG on the nucleation program, HMS secondary polymerized to crystalline SiO_2 through the edge dehydration condensation process, indicating that heavy metals enter the interior of the soil lattice by participating in the isomorphous replacement (Fig. 6E, F). Subsequently, with the progress of MCG degradation and microbial activities, the soil gradually returned to neutrality. AHMS in the soil entered the second sealing pathway (**Eq. S14-S17**), forming numerous new soil lattices with central atoms replaced by heavy metals, which verified the catalytic effect of “electron cage” structures on the formation process of HMS (**Eq. S16-S17**). In addition, the distribution of heavy metals in the surface phase should theoretically decrease with the progress of the NS mechanism. Normalization can greatly clarify the properties of spectral differences and simplify the interpretation of reflection features in spectral images [67]. Therefore, the surface phase characterization technique (FT-IR) was used to detect the vibration modes of heavy metals at different pH levels and to perform a normalization [68]. The results showed that, with the decrease in pH, the characteristic peak of the Si—O bond was blue-shifted due to the polymerization of HMS. Meanwhile, the characteristic peak of heavy metal gradually weakened until it disappeared (Fig. 6G, H), indicating that the heavy metal successfully entered the interior of the soil lattice through the isomorphic displacement effect. Consistent with this, the size distribution of soil particles increased from 0.002 to 0.020 mm to 0.200–2.000 mm with the development of nucleation mechanism (**Fig. S9**), which was attributed to the promoting effect of MCG on HMS formation and polymerization through the “electron cage” effect.

More excitingly, after 90 days of aging and microbial activity, although pH and σ have returned to the original soil level (Fig. 6D), passivation rates remain at extremely high levels (Pb: 99.96 %, Cd: 99.93 %, Zn: 99.71 %), consistent with the thermodynamic irreversibility of the NS mechanism described in the above calculations. To further verify the stability of heavy metals inside the lattice and analyze the distribution depth of different heavy metals, XPS combined etching technology was used to etch the soil after 90 days of aging from the surface to the interior of the lattice (Time range: 2 min) and detect the distribution of heavy metals (Fig. 6I, J). The results show that Pb ions are mainly concentrated in the deep layer of the crystal, and the concentration of Cd ions increases steadily with etching and distributes evenly in different depths. Meanwhile, Zn ions are mainly distributed in the surface layer of the bulk phase, which is completely consistent with the nucleation tendency of ΔG in Fig. 1D (Pb < Cd < Zn). It was proved that MCG successfully completed soil isomorphic replacement by inducing the soil NS mechanism.

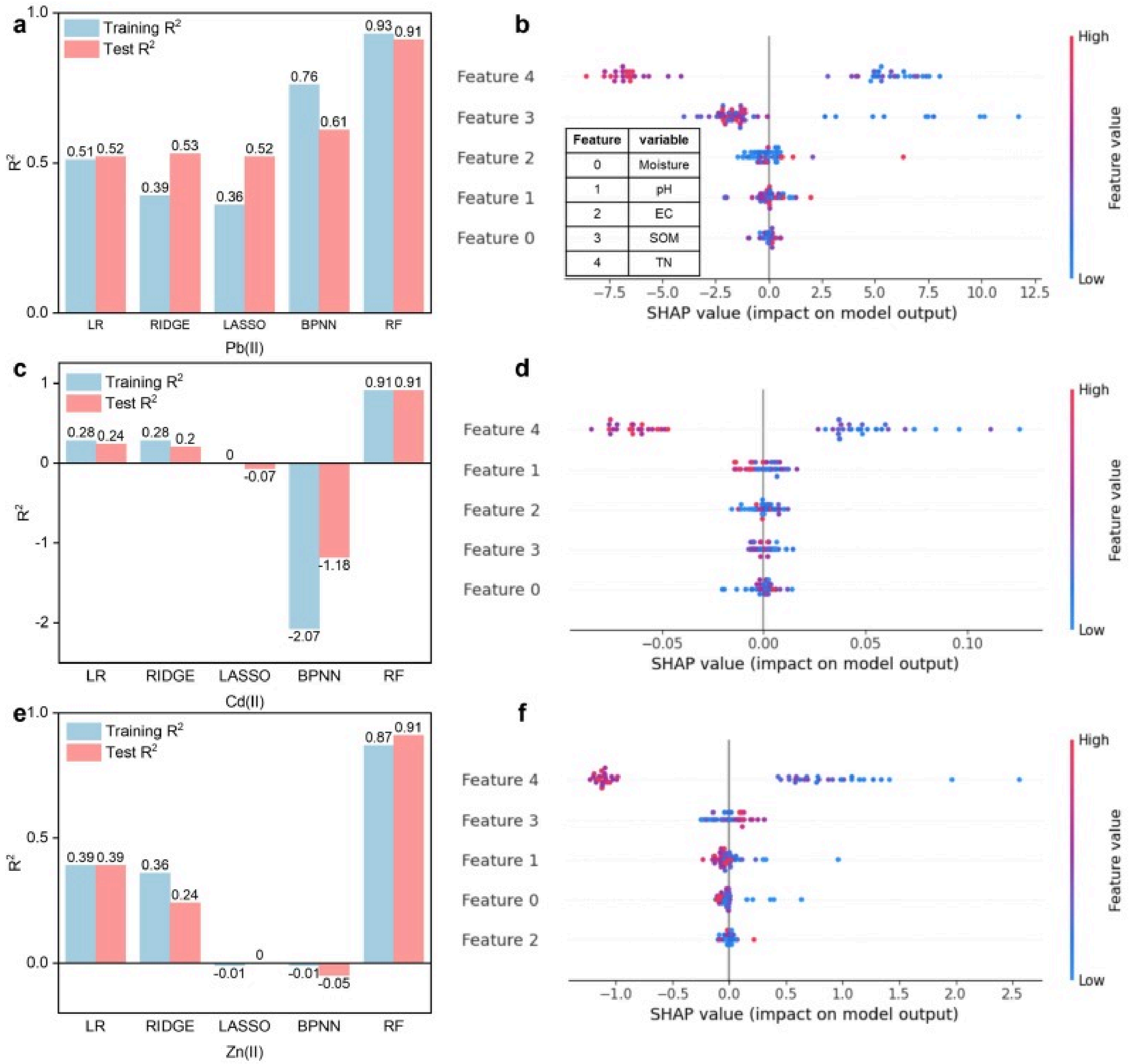
In summary, MCG used the special “O-N-O" electron cage effect of CS to induce soil isomorphic replacement, sealing heavy metals inside the soil lattice for a long time, breaking through the material dependence of traditional remediation schemes, and showing passivation efficiency and stability far beyond the traditional passivation mechanisms (complex, precipitation, ion exchange, and electrostatic attraction).

3.9. Evaluation of physical and chemical properties of soil

In addition to remediation, improving soil quality (such as carbon sink and fertility) is the development direction of advanced environmental functional materials [69]. Therefore, we supplemented the evaluation of the effects of MCG on basic soil functions when sealing heavy metals by monitoring soil conductivity, pH, soil organic carbon (SOC), and total nitrogen (TN) for 60 days after remediation (Fig. 6K). At 60 days, soil electrical conductivity decreased to 2.14 ± 0.17 mS·cm^{-1} (original soil: 2.23 ± 0.14 mS·cm^{-1}) and pH increased to 6.90 ± 0.35 (original soil: 6.47 ± 0.16), indicating that soil resilience was not damaged by MCG, and soil microenvironment balance could be quickly restored after degradation. The soil electrical conductivity returned to the background level on the 13_{th} day and coincided with the peak of immobilization efficiency. This phenomenon dynamically coupled the essence of ion migration and fixation during the remediation process: initially, the alkaline hydrolysis effect of MCG caused the release of ions (including silicates and heavy metal ions) from soil minerals, resulting in an increase in conductivity; subsequently, its “electron cage” structure catalyzed the rapid conversion of both into stable heavy metal silicates and their consolidation in the soil lattice through the nucleation-sealing mechanism, which consumed a large amount of free ions in the solution, causing the conductivity to drop; when the ion sealing reaction was nearly completed, the conductivity returned to a steady state determined by the soil's own properties, marking the irreversible fixation of heavy metals and the cessation of the reliance on external environmental ion concentrations, thus achieving the unity of efficiency and environmental compatibility. In addition, soil organic carbon (SOC) increased from 26.40 ± 1.47 g·Kg^{-1} to 47.70 ± 2.09 g·Kg^{-1}, and total nitrogen (TN) increased from 1.46 ± 0.57 g·Kg^{-1} to 1.94 ± 0.33 g·Kg^{-1}. Our results suggested that MCG, composed of urea, cellulose, and chitosan, could not only induce long-term sealing of heavy metals in the soil through the mechanism of NS, but also effectively improve soil fertility based on low environmental disturbance. In summary, MCG is an ideal advanced remediation material for heavy metal-contaminated soil.

3.10. Actual contaminated site remediation

To verify that MCG can still achieve ideal remediation performance through the NS mechanism at actual contaminated sites, MCG was applied according to the optimal application conditions to 50 typical heavy metal-contaminated sites in Jinzhou City, Liaoning Province, China. After being cultivated in the actual environment for 30days, soil samples after remediation were collected, and a series of tests were conducted on their electrical conductivity (EC), moisture content, pH, soil organic matter content (SOM), total nitrogen (TN), and bioavailable Pb(II), Cd(II), and Zn(II). The data (Table S4 were comprehensively analyzed using multiple linear regression (LR), ridge regression (RIDGE), lasso regression (LASSO), backpropagation neural network (BPNN), and random forest (RF) models, as well as Pearson correlation analysis (**Text S8**). Furthermore, SHapley Additive exPlanations (SHAP) was employed to conduct a mechanism and contribution analysis on the RF fitting results. The fitting results of different models are shown in Fig. 7A, C and E. For both bioavailable Pb(II), Cd(II) and Zn(II), the RF model has the best fitting degree (coefficient of determination, R^2) and generalization ability (**Fig. S10**). Its R^2 values are all close to or exceed 0.9, indicating that the RF model can accurately predict the bioavailable content of heavy metals based on soil physical and chemical properties. Thus, the RF model is highly reliable and an excellent tool for predicting the bioavailability of heavy metals, providing a key basis for establishing rapid and accurate soil remediation effect prediction models in the future. The Pearson correlation analysis (**Fig. S11**) indicated that TN and SOM were significantly negatively correlated with the bioavailable content of heavy metals. Interestingly, TN was unexpectedly positively correlated with pH (in traditional theory, TN and pH are usually negatively correlated), which precisely indicates that the reduction of heavy metal availability is dominated by TN, and the change of TN is mainly controlled by the alkaline CS. More importantly, in Fig. 7B, D and F, through the SHAP value analysis, we revealed the contribution degree and influence direction of different soil physical and chemical properties of the model prediction results (i.e., the bioavailable contents of different heavy metals), with the color intensity representing the magnitude of the characteristic values. The results show that, except for EC (the inherent property of the soil), MCG jointly controls the contents of heavy metal availability through the two key factors (SOM and TN), confirming that during the remediation process, the NS mechanism and the complexation with organic matter occur in synergy, and the nucleation mechanism represented by TN occupies an absolute dominant position.



Feature	variable
0	Moisture
1	pH
2	EC
3	SOM
4	TN

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Fig. 7. **Machine learning to decode remediation data from contaminated sites.** The degree of fit (**A**) and interpretability analysis (**B**) of different models to soil physical and chemical properties and bioavailable Pb(II) content; The degree of fit (**C**) and

interpretability analysis (**D**) of different models for soil physical and chemical properties and bioavailable Cd(II) content; Degree of fit (**E**) and interpretability analysis (**F**) of different models for soil physical and chemical properties and bioavailable Zn(II) content.

4. Discussions

Soil organic matter interface processes have profound implications for both natural and engineered systems. Under the catalysis of organic matter with strong electrostatic interaction (such as chitosan), heavy metals can be immobilized irreversibly within the lattices of soils via isomorphous replacement. In this study, we first revealed the “electron cage” effect of the special structure-activity relationship of chitosan through first-principles calculation combined with characterization technology. Based on this, we proposed a strategy to achieve long-term fixation of heavy metals (Pb, Zn, Cd) in the soil lattice by activating the soil intrinsic isomorphic replacement mechanism. Different from traditional remediation schemes that rely on the input of exogenous materials, the microporous chitosan/cellulose self-assembled biogels (MCG) (prepared by the freeze-thaw cycle method with gradient cooling) can drive the soil to “autonomously” seal heavy metals inside the lattice through a dual-path nucleation mechanism. This innovation not only achieves extremely high passivation efficiency (>99.7% reduction in bioavailability), but also improves soil fertility without damaging soil resilience, providing a breakthrough solution for sustainable soil remediation.

Impressively, the irreversible fixation ability of MCG results from the breakthrough of two key thermodynamic and kinetic bottlenecks: (1) Spontaneous de-alkalization of alkaline heavy metal silicates at high pH; (2) Acceleration of the slow nucleation process of heavy metal silicates. Using density functional theory calculations, we reveal that the above two breakthroughs result from the catalysis of chitosan's unique “O-N-O" electron cage structure. The HMS nuclear barrier is exponentially reduced by pinching the excessive electron transfer of the amino group and polarizing the heavy metal ions in $Si(OH)_3OM^+$, thus transforming the traditional inert isomorphous replacement process into a fast, spontaneous and irreversible process. This discovery clarified the mechanism of functional groups regulating the evolution of soil minerals at the atomic level and provided a new theoretical perspective for the field of environmental chemistry.

It is worth noting that MCG achieves supramolecular self-assembly of CS and MCC through the FTC strategy of gradient cooling, which can form a multistage pore structure (without a crosslinking agent) and alkaline surface. This design maximizes the exposure of the "electron cage" active sites, enabling MCG to dissolve the soil, catalyze HMS formation, and degrade harmlessly after remediation is complete. A 90-day simulated field experiment further confirmed that the remediation effect of MCG is thermodynamic irreversibility. The XPS-etching technique showed that the heavy metal is deeply locked in the soil lattice, and even if the MCG degrades, there is no significant rebound in bioavailability. In addition, MCG can significantly increase soil organic carbon and total nitrogen while sealing heavy metals, becoming one of the few materials with both controlling pollution and restoring soil physicochemical properties. In addition, the evolution of major cations and anions in soil is a worth study in the future to reveal the regulation of soil mineral composition by organic matter. Meanwhile, it is also necessary to assess how the solid phase distribution of target heavy metals changes after application of bio-gels [70,71].

In summary, this study offers a novel approach to developing heavy metal remediation materials by shifting from material-focused soil remediation to material-assisted soil self-remediation, while providing new insights into the environmental transformation of heavy metals in soils.

5. Conclusions

In summary, this study clarified the potential pathways for heavy metal edge nucleation in various soil environments, revealed the catalytic mechanism of the "electron cage" structure on HMS nucleation, and identified a new pathway for spontaneous AHMS de-alkalization enabled by the "electron cage". MCG with alkaline surface, abundant pores and "electron cage" structures was synthesized by FTC with gradient cooling, and exponentially accelerated the isomorphous replacement process at soil edges, overcoming the thermodynamic limitations of conventional passivation, achieving >99.7% reduction in bioavailability of Pb, Cd, and Zn within 13 days while enhancing soil fertility (SOC increased by 81%, TN increased by 33%). Crucially, MCG's supramolecular self-assembly structure, engineered without crosslinking agents via gradient-cooling freeze-thaw cycles, maximizes exposure of the catalytic “O-N-O" sites, ensuring minimal environmental disturbance. In summary, this work establishes a material-driven soil self-healing paradigm, shifting remediation from exogenous input dependency to intrinsic lattice-sealing activation, with broad implications for sustainable soil management under SDG 15.

CRediT authorship contribution statement

Gaoyuan Gu: Writing – original draft, Software, Investigation, Formal analysis, Data curation, Conceptualization. **Yan Zhou:** Visualization, Validation, Formal analysis. **Jianing Zhang:** Writing – review & editing, Methodology. **Xinyao Wang:** Visualization, Investigation. **Chong Peng:** Writing – review & editing, Supervision, Software, Resources. **Yuanfei Wang:** Resources, Project administration. **Changlong Bi:** Writing – review & editing. **Shuyi Yang:** Writing – review & editing, Funding acquisition, Conceptualization. **Tao E:** Writing – review & editing, Visualization, Project administration, Funding acquisition, Conceptualization.

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Liaoning Province (Jinzhou) Fur Green Manufacturing Industry Technology Innovation Strategic Alliance [201854] Research on efficient control on the crystal surface of titanium dioxide to prepare nano-conductive materials and its key technologies for industrialization [XLYC2403193] Research on the coupling mechanism of soil biological fertilization and carbon sequestration based on big data analysis on the Bohai environmental sample bank (2025JH2/101300070).

PhD launch project of Bohai University: Research on the mobility of chromium from tannery sludge waste and its mechanism on graded-stabilization [0523BS056].

Declaration of competing interest

Authors declare that they have no competing interests.

Appendix A. Supplementary data

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Supplementary material

Data availability

Data will be made available on request.

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